

Chapter 1

Ideation Phase

1.1 Brainstorm & Idea Prioritization Template

Date: 7 juy 2026

Project Name: Book Store: Online Book Shopping System Using MERN Stack

Brainstorming provides a free and open environment that encourages every team member to participate in creative thinking and problem solving. During the brainstorming session, several ideas were discussed to identify a practical and innovative project. After evaluating different concepts based on feasibility, usability, and implementation complexity, the team selected the Online Book Store project.

The brainstorming activity helped in identifying the key features required for developing a secure, scalable, and user-friendly online bookstore using the MERN Stack.

1.2 Step 1: Team Gathering, Collaboration and Selection of Problem Statement

The project team organized a brainstorming session to discuss various ideas related to web application development. Several project ideas were proposed, including an Online Library Management System, Student Management System, Food Ordering System, E-Learning Platform, and Online Book Store.

After analyzing the advantages, technical feasibility, and user requirements, the team finalized the following problem statement.

Selected Problem Statement

Develop a secure and user-friendly Online Book Store application using the MERN Stack that enables users to browse books, search by category, maintain wishlists, manage shopping carts, place orders, and provide reviews and ratings while allowing administrators to manage books, users, and customer orders efficiently.

1.3 Step 2: Brainstorm, Idea Listing and Grouping

The following ideas were generated during the brainstorming session.

Table 1.1: Brainstormed Ideas

S.No	Idea	Category
1	User Registration and Login	Authentication
2	JWT Authentication	Security
3	Browse Books	Book Management
4	Search Books	Search Module
5	Category-wise Books	Catalog
6	Shopping Cart	E-Commerce
7	Wishlist	User Features
8	Reviews and Ratings	Customer Feedback
9	Order Management	Transactions
10	Admin Dashboard	Administration
11	Inventory Management	Administration
12	Responsive User Interface	Frontend
13	Password Encryption using bcrypt	Security
14	REST API Communication	Backend
15	MongoDB Database Integration	Database

Idea Grouping

The generated ideas were grouped into the following categories.

- User Management

- User Registration
 - User Login
 - JWT Authentication
 - User Profile
- **Book Management**
 - Browse Books
 - Search Books
 - Categories
 - Book Details
- **Shopping Features**
 - Shopping Cart
 - Wishlist
 - Orders
- **Customer Interaction**
 - Reviews
 - Ratings
 - Order History
- **Administration**
 - Admin Dashboard
 - Inventory Management
 - User Management
 - Order Management

1.4 Step 3: Idea Prioritization

The identified ideas were prioritized based on their importance, technical feasibility, and value to end users.

Table 1.2: Idea Prioritization

Feature	Importance	Feasibility	Priority
User Registration and Login	High	High	High
JWT Authentication	High	High	High
Book Catalog	High	High	High
Search Books	High	High	High
Shopping Cart	High	High	High
Wishlist	High	High	High
Reviews and Ratings	Medium	High	Medium
Order Management	High	High	High
Admin Dashboard	High	Medium	High
Inventory Management	Medium	High	Medium
Email Notifications	Medium	Medium	Low
AI-Based Recommendations	Low	Low	Low

Final Prioritized Features

The following features were selected for implementation in the first version of the Book Store application.

1. User Registration and Login
2. JWT Authentication
3. Book Catalog
4. Category-wise Browsing
5. Search Functionality
6. Shopping Cart
7. Wishlist
8. Reviews and Ratings
9. Order Management
10. Admin Dashboard

1.5 Outcome of the Ideation Phase

The brainstorming and idea prioritization process enabled the team to identify the most important features required for the Book Store application. By focusing on secure authentication, efficient book management, shopping cart functionality, wishlist, reviews, ratings, and order management, the team established a clear development roadmap. The selected features ensure that the application is scalable, secure, and provides a seamless online shopping experience for users while simplifying administrative tasks.